Chapter 3

4. Since a 4-digit code can have between zero and four 0s, the possible values of X are 0, 1, 2, 3, 4. As examples, the PIN 9876 yields $X = 0$ (no 0s), 1006 corresponds to $X = 2$ (two 0s), and the very poor PIN choice of 0000 implies that $X = 4$.

6. The possible X values are1, 2, 3, 4, … (all positive integers). Some examples are:

Outcome:	*RL*	*AL*	*RAARL*	*RRRRL*	*AARRL*
X:	2	2	5	5	5

10.

a. Possible values of T are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

b. Possible values of X are: –4, –3, –2, –1, 0, 1, 2, 3, 4, 5, 6.

c. Possible values of U are: 0, 1, 2, 3, 4, 5, 6.

d. Possible values of Z are: 0, 1, 2.

12.

a. Since there are 50 seats, the flight will accommodate all ticketed passengers who show up as long as there are no more than 50. $P(Y \leq 50) = .05 + .10 + .12 + .14 + .25 + .17 = .83$.

b. This is the complement of part **a**: $P(Y > 50) = 1 - P(Y \leq 50) = 1 - .83 = .17$.

c. If you're the first standby passenger, you need no more than 49 people to show up (so that there's space left for you). $P(Y \leq 49) = .05 + .10 + .12 + .14 + .25 = .66$. On the other hand, if you're <u>third</u> on the standby list, you need no more than 47 people to show up (so that, even with the two standby passengers ahead of you, there's still room). $P(Y \leq 47) = .05 + .10 + .12 = .27$.

14.

a. As the hint indicates, the sum of the probabilities must equal 1. Applied here, we get $\sum_{y=1}^{5} p(y) = k[1+2+3+4+5] = 15k = 1 \Rightarrow k = \frac{1}{15}$. In other words, the probabilities of the five y-values are $\frac{1}{15}, \frac{2}{15}, \frac{3}{15}, \frac{4}{15}, \frac{5}{15}$.

b. $P(Y \le 3) = P(Y = 1, 2, 3) = \frac{1}{15} + \frac{2}{15} + \frac{3}{15} = \frac{6}{15} = .4$.

c. $P(2 \le Y \le 4) = P(Y = 2, 3, 4) = \frac{2}{15} + \frac{3}{15} + \frac{4}{15} = \frac{9}{15} = .6$.

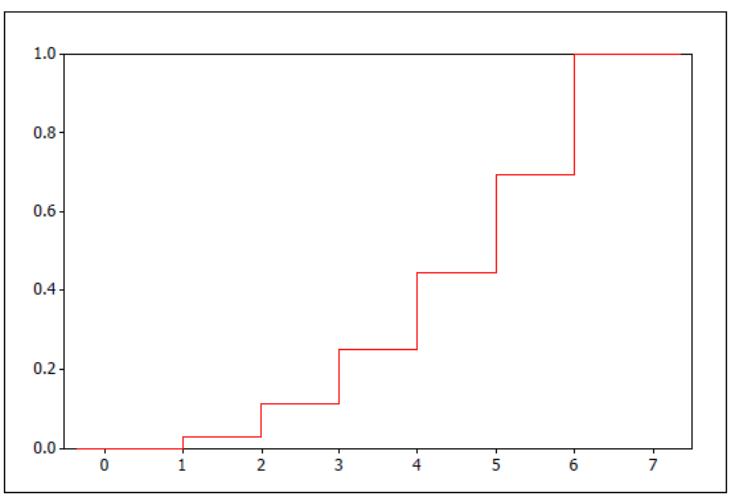
d. Do the probabilities total 1? Let's check: $\sum_{y=1}^{5}\left(\frac{y^2}{50}\right) = \frac{1}{50}[1+4+9+16+25] = \frac{55}{50} \ne 1$. No, that formula cannot be a pmf.

18.

a. $p(1) = P(M = 1) = P(\{(1,1)\}) = \frac{1}{36}$; $p(2) = P(M = 2) = P(\{(1,2)(2,1)(2,2)\}) = \frac{3}{36}$; $p(3) = P(M = 3) = P(\{(1,3)(2,3)(3,1)(3,2)(3,3)\}) = \frac{5}{36}$. Continuing the pattern, $p(4) = \frac{7}{36}$, $p(5) = \frac{9}{36}$, and $p(6) = \frac{11}{36}$.

b. Using the values in **a**,

$$F(m) = \begin{cases} 0 & m < 1 \\ \frac{1}{36} & 1 \le m < 2 \\ \frac{4}{36} & 2 \le m < 3 \\ \frac{9}{36} & 3 \le m < 4 \\ \frac{16}{36} & 4 \le m < 5 \\ \frac{25}{36} & 5 \le m < 6 \\ 1 & m \ge 6 \end{cases}$$



28. If $x_1 < x_2$, then $F(x_2) = P(X \le x_2) = P(\{X \le x_1\} \cup \{x_1 < X \le x_2\}) = P(X \le x_1) + P(x_1 < X \le x_2)$. Since all probabilities are non-negative, this is $\ge P(X \le x_1) + 0 = P(X \le x_1) = F(x_1)$. That is, $x_1 < x_2$ implies $F(x_1) \le F(x_2)$, QED.

Looking at the proof above, $F(x_1) = F(x_2)$ iff $P(x_1 < X \le x_2) = 0$.

32.

a. $E(X) = (16)(.2) + (18)(.5) + (20)(.3) = 18.2$ ft^3; $E(X^2) = (16)^2(.2) + (18)^2(.5) + (20)^2(.3) = 333.2$. Put these together, and $V(X) = E(X^2) - [E(X)]^2 = 333.2 - (18.2)^2 = 1.96$.

b. Use the linearity/rescaling property: $E(70X - 650) = 70\mu - 650 = 70(18.2) - 650 = \$624$. Alternatively, you can figure out the price for each of the three freezer types and take the weighted average.

c. Use the linearity/rescaling property again: $V(70X - 650) = 70^2\sigma^2 = 70^2(1.96) = 9604$. (The 70 gets squared because variance is itself a square quantity.)

d. We cannot use the rescaling properties for $E(X - .008X^2)$, since this isn't a linear function of X. However, since we've already found both $E(X)$ and $E(X^2)$, we may as well use them: the expected actual capacity of a freezer is $E(X - .008X^2) = E(X) - .008E(X^2) = 18.2 - .008(333.2) = 15.5344$ ft^3. Alternatively, you can figure out the actual capacity for each of the three freezer types and take the weighted average.

36. You have to be careful here: if \$0 damage is incurred, then there's no deductible for the insured driver to pay! Here's one approach: let $h(X)$ = the amount paid by the insurance company on an accident claim, which is \$0 for a "no damage" event and \$500 less than actual damages ($X - 500$) otherwise. The pmf of $h(X)$ looks like this:

x	0	1000	5000	10000
$h(x)$	0	500	4500	9500
$p(x)$	.8	.1	.08	.02

Based on the pmf, the average payout across these types of accidents is $E(h(X)) = 0(.8) + 500(.1) + 4500(.08) + 9500(.02) = \$600$. If the insurance company charged \$600 per client, they'd break even (a bad idea!). To have an expected profit of \$100 — that is, to have a *mean* profit of \$100 per client — they should charge \$600 + \$100 = \$700.

42.

a. $E[X(X-1)] = E(X^2) - E(X) \Rightarrow E(X^2) = E[X(X-1)] + E(X) = 27.5 + 5 = 32.5$.

b. $V(X) = E(X^2) - [E(X)]^2 = 32.5 - (5)^2 = 7.5$.

c. Substituting **a** into **b**, $V(X) = E[X(X-1)] + E(X) - [E(X)]^2$.

48. $X \sim \text{Bin}(25, .05)$

a. $P(X \le 3) = B(3;25,.05) = .966$, while $P(X < 3) = P(X \le 2) = B(2;25,.05) = .873$.

b. $P(X \ge 4) = 1 - P(X \le 3) = 1 - B(3;25,.05) = .1 - .966 = .034$.

c. $P(1 \le X \le 3) = P(X \le 3) - P(X \le 0) = .966 - .277 = .689$.

d. $E(X) = np = (25)(.05) = 1.25$, $\sigma_X = \sqrt{np(1-p)} = \sqrt{25(.05)(.95)} = 1.09$.

e. With $n = 50$, $P(X = 0) = \binom{50}{0}(.05)^0(.95)^{50} = (.95)^{50} = .077$.

50. Let X be the number of faxes, so $X \sim \text{Bin}(25, .25)$.

a. $P(X \le 6) = B(6;25,.25) = .561$.

b. $P(X = 6) = b(6;25,.25) = .183$.

c. $P(X \ge 6) = 1 - P(X \le 5) = 1 - B(5;25,.25) = .622$.

d. $P(X > 6) = 1 - P(X \le 6) = 1 - .561 = .439$.

52. Let X be the number of students who want a new copy, so $X \sim \text{Bin}(n = 25, p = .3)$.

a. $E(X) = np = 25(.3) = 7.5$ and $SD(X) = \sqrt{np(1-p)} = \sqrt{25(.3)(.7)} = 2.29$.

b. Two standard deviations from the mean converts to $7.5 \pm 2(2.29) = 2.92$ & 12.08. For X to be <u>more</u> than two standard deviations from the means requires $X < 2.92$ or $X > 12.08$. Since X must be a non-negative integer, $P(X < 2.92 \text{ or } X > 12.08) = 1 - P(2.92 \le X \le 12.08) = 1 - P(3 \le X \le 12) =$

$$1 - \sum_{x=3}^{12} \binom{25}{x} (.3)^x (.7)^{25-x} = 1 - .9736 = .0264.$$

c. If $X > 15$, then more people want new copies than the bookstore carries. At the other end, though, there are $25 - X$ students wanting used copies; if $25 - X > 15$, then there aren't enough used copies to meet demand.

The inequality $25 - X > 15$ is the same as $X < 10$, so the bookstore can't meet demand if either $X > 15$ or $X < 10$. All 25 students get the type they want iff $10 \le X \le 15$:

$$P(10 \le X \le 15) = \sum_{x=10}^{15} \binom{25}{x} (.3)^x (.7)^{25-x} = .1890.$$

d. The bookstore sells X new books and $25 - X$ used books, so total revenue from these 25 sales is given by $h(X) = 100(X) + 70(25 - X) = 30X + 1750$. Using linearity/rescaling properties, expected revenue equals $E(h(X)) = E(30X + 1750) = 30\mu + 1750 = 30(7.5) + 1750 = \$1975$.

54. Let X equal the number of customers who choose an oversize racket, so $X \sim \text{Bin}(10, .60)$.

a. $P(X \ge 6) = 1 - P(X \le 5) = 1 - B(5;20,.60) = 1 - .367 = .633$.

b. $\mu = np = 10(.6) = 6$ and $\sigma = \sqrt{10(.6)(.4)} = 1.55$, so $\mu \pm \sigma = (4.45, 7.55)$.
$P(4.45 < X < 7.55) = P(5 \le X \le 7) = P(X \le 7) - P(X \le 4) = .833 - .166 = .667$.

c. This occurs iff between 3 and 7 customers want the oversize racket (otherwise, one type will run out early). $P(3 \le X \le 7) = P(X \le 7) - P(X \le 2) = .833 - .012 = .821$.

56. Let X = the number of students in the sample needing accommodation, so $X \sim \text{Bin}(25, .02)$.

a. $P(X = 1) = 25(.02)(.98)^{24} = .308$.

b. $P(X \ge 1) = 1 - P(X=0) = 1 - (.98)^{25} = 1 - .603 = .397$.

c. $P(X \ge 2) = 1 - P(X \le 1) = 1 - [.603 + .308] = .089$.

d. $\mu = 25(.02) = .5$ and $\sigma = \sqrt{25(.02)(.98)} = \sqrt{.49} = .7$, so $\mu \pm \sigma = (-.9, 1.9)$. $P(-.9 \le X \le 1.9) =$ $P(X = 0 \text{ or } 1) = .911$.

e. $\dfrac{.5(4.5) + 24.5(3)}{25} = 3.03$ hours. Notice the sample size of 25 is actually irrelevant.

60. Using the hint, $h(X) = 1 \cdot X + 2.25(25 - X) = 62.5 - 1.5X$, which is a linear function. Since the mean of X is $E(X) = np = (25)(.6) = 15$, $E(h(X)) = 62.5 - 1.5E(X) = 62.5 - 1.5(15) = \$40$.

66.

a. Let Y = the number with reservations who show up, a binomial rv with $n = 6$ and $p = .8$. Since there are only four spaces available, at least one individual cannot be accommodated if Y is more than 4. The desired probability is $P(Y = 5 \text{ or } 6) = b(5; 6, .8) + b(6;6,.8) = .3932 + .2621 = .6553$.

b. Let $h(Y)$ = the number of available spaces. Then

when y is:	0	1	2	3	4	5	6
$h(y)$ is:	4	3	2	1	0	0	0

The expected number of available spaces when the limousine departs equals

$$E[h(Y)] = \sum_{y=0}^{6} h(y) \cdot b(y;6,.8) = 4(.000) + 3(.002) = 2(.015) + 1(.082) + 0 + 0 + 0 = 0.118.$$

c. Possible X values are 0, 1, 2, 3, and 4. $X = 0$ if there are 3 reservations and 0 show or 4 reservations and 0 show or 5 reservations and 0 show or 6 reservations and none show, so
$P(X = 0) = b(0; 3, .8)(.1) + b(0; 4, .8)(.2) + b(0; 5, .8)(.3) + b(0; 6, .8)(.4)$
$= .0080(.1) + .0016(.2) + .0003(.3) + .0001(.4) = .0013$.
Similarly, $P(X = 1) = b(1; 3, .8)(.1) + \ldots + b(1; 6, .8)(.4) = .0172$; $P(X = 2) = .0906$;
$P(X = 3) = .2273$; and $P(X = 4) = .6636$.

These values are displayed below.

x	0	1	2	3	4
$p(x)$	.0013	.0172	.0906	.2273	.6636

68.

a. There are 18 items (people) total, 8 of which are "successes" (first-time examinees). Among these 18 items, 6 have been randomly assigned to this particular examiner. So, the random variable X is hypergeometric, with $N = 18$, $M = 8$, and $n = 6$.

b. $$P(X = 2) = \frac{\binom{8}{2}\binom{18-8}{6-2}}{\binom{18}{6}} = \frac{\binom{8}{2}\binom{10}{4}}{\binom{18}{6}} = \frac{(28)(210)}{18564} = .3167.$$

$$P(X \le 2) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{\binom{8}{0}\binom{10}{6}}{\binom{18}{6}} + \frac{\binom{8}{1}\binom{10}{5}}{\binom{18}{6}} + .3167 =$$

$.0113 + .1086 + .3167 = .4366$.
$P(X \ge 2) = 1 - P(X \le 1) = 1 - [P(X = 0) + P(X = 1)] = 1 - [.0113 + .1086] = .8801$.

c. $E(X) = n \cdot \frac{M}{N} = 6 \cdot \frac{8}{18} = 2.67$; $V(X) = \left(\frac{18-6}{18-1}\right) \cdot 6\left(\frac{8}{18}\right)\left(1 - \frac{8}{18}\right) = 1.04575$; $\sigma = 1.023$.

72.

a. There are $N = 11$ candidates, $M = 4$ in the "top four" (obviously), and $n = 6$ selected for the first day's interviews. So, the probability x of the "top four" are interviewed on the first day equals $h(x; 6, 4, 11) =$

$$\frac{\binom{4}{x}\binom{7}{6-x}}{\binom{11}{6}}.$$

b. With X = the number of "top four" interview candidates on the first day, $E(X) = n \cdot \frac{M}{N} = 6 \cdot \frac{4}{11} = 2.18$.

76. This question relates to the negative binomial distribution, but we can't use it directly (X, as it's defined, doesn't fit the negative binomial description). Instead, let's reason it out.
Clearly, X can't be 0, 1, or 2.
$P(X = 3) = P(BBB \text{ or } GGG) = (.5)^3 + (.5)^3 = .25$.
$P(X = 4) = P(BBGB \text{ or } BGBB \text{ or } GBBB \text{ or } GGBG \text{ or } GBGG \text{ or } BGGG) = 6(.5)^4 = .375$.
$P(X = 5)$ look scary until we realize that 5 is the maximum possible value of X!
(If you have 5 kids, you must have at least 3 of one gender.) So, $P(X = 5) = 1 - .25 - .375 = .375$, and that completes the pmf of X.

x	3	4	5
$p(x)$	.25	.375	.375

80. Solutions are found using the cumulative Poisson table, $F(x; \mu) = F(x; 4)$.

a. $P(X \le 4) = F(4; 4) = .629$, while $P(X < 4) = P(X \le 3) = F(3; 4) = .434$.

b. $P(4 \le X \le 8) = F(8; 4) - F(3; 4) = .545$.

c. $P(X \ge 8) = 1 - P(X < 8) = 1 - P(X \le 7) = 1 - F(7; 4) = .051$.

d. For this Poisson model, $\mu = 4$ and so $\sigma = \sqrt{4} = 2$. The desired probability is $P(X \le \mu + \sigma) = P(X \le 4 + 2) = P(X \le 6) = F(6; 4) = .889$.

82.

a. $P(X = 1) = F(1; .2) - F(0; .2) = .982 - .819 = .163$.

b. $P(X \ge 2) = 1 - P(X \le 1) = 1 - F(1; .2) = 1 - .982 = .018$.

c. The probability a disk contains zero missing pulses is $P(X = 0) = \frac{e^{-2}(.2)^0}{0!} = .819$. Since the two disks are independent, $P(1^{\text{st}} \text{ doesn't} \cap 2^{\text{nd}} \text{ doesn't}) = P(1^{\text{st}} \text{ doesn't}) \cdot P(2^{\text{nd}} \text{ doesn't})$
$= (.819)(.819) = .671$.

88. Let X = the number of diodes on a board that fail. Then $X \sim \text{Bin}(n = 200, p = .01)$.

a. $E(X) = np = (200)(.01) = 2$; $V(X) = npq = (200)(.01)(.99) = 1.98$, so $\sigma = 1.407$.

b. X has approximately a Poisson distribution with $\mu = np = 2$, so $P(X \geq 4) = 1 - P(X \leq 3) = 1 - F(3; 2) = 1 - .857 = .143$.

c. For any one board, P(board works properly) = P(all diodes work) = $P(X = 0) = \dfrac{e^{-2}2^0}{0!} = .135$.

Let Y = the number among the five boards that work, a binomial rv with $n = 5$ and $p = .135$.

Then $P(Y \geq 4) = P(Y = 4) + P(Y = 5) = \binom{5}{4}(.135)^4(.865) + \binom{5}{5}(.135)^5(.865)^0 = .00148$.

Chapter 4

2. $f(x) = \frac{1}{10}$ for $-5 \leq x \leq 5$ and $= 0$ otherwise

a. $P(X < 0) = \int_{-5}^{0} \frac{1}{10}\,dx = .5$.

b. $P(-2.5 < X < 2.5) = \int_{-2.5}^{2.5} \frac{1}{10}\,dx = .5$.

c. $P(-2 \leq X \leq 3) = \int_{-2}^{3} \frac{1}{10}\,dx = .5$.

d. $P(k < X < k + 4) = \int_{k}^{k+4} \frac{1}{10}\,dx = \frac{1}{10}x\Big]_{k}^{k+4} = \frac{1}{10}[(k+4) - k] = .4$.

4.

a. $\int_{-\infty}^{\infty} f(x;\theta)dx = \int_0^{\infty} \frac{x}{\theta^2} e^{-x^2/2\theta^2}\,dx = -e^{-x^2/2\theta^2}\Big]_0^{\infty} = 0 - (-1) = 1$

b. $P(X \leq 200) = \int_{-\infty}^{200} f(x;\theta)dx = \int_0^{200} \frac{x}{\theta^2} e^{-x^2/2\theta^2}\,dx = -e^{-x^2/2\theta^2}\Big]_0^{200} \approx -.1353 + 1 = .8647$.

$P(X < 200) = P(X \leq 200) \approx .8647$, since X is continuous.

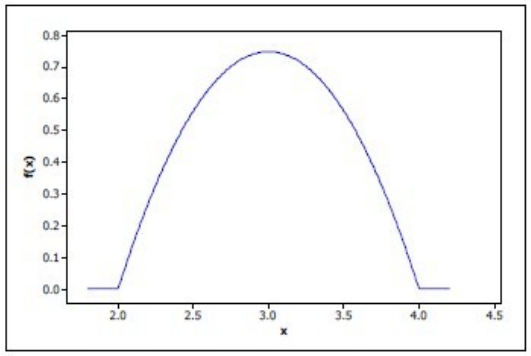
$P(X \geq 200) = 1 - P(X < 200) \approx .1353$.

c. $P(100 \leq X \leq 200) = \int_{100}^{200} f(x;\theta)dx = -e^{-x^2/20{,}000}\Big]_{100}^{200} \approx .4712$.

d. For $x > 0$, $P(X \leq x) = \int_{-\infty}^{x} f(y;\theta)dy = \int_0^x \frac{y}{\theta^2} e^{-y^2/2\theta^2}\,dx = -e^{-y^2/2\theta^2}\Big]_0^x = 1 - e^{-x^2/2\theta^2}$.

6.

a.



b. $1 = \int_2^4 k[1-(x-3)^2]dx = \int_{-1}^1 k[1-u^2]du = \cdots = \dfrac{4k}{3} \Rightarrow k = \dfrac{3}{4}$.

c. $P(X > 3) = \int_3^4 \frac{3}{4}[1-(x-3)^2]dx = .5$. This matches the symmetry of the pdf about $x = 3$.

d. $P\left(\frac{11}{4} \le X \le \frac{13}{4}\right) = \int_{11/4}^{13/4} \frac{3}{4}[1-(x-3)^2]dx = \frac{3}{4}\int_{-1/4}^{1/4}[1-u^2]du = \dfrac{47}{128} \approx .367$.

e. $P(|X-3| > .5) = 1 - P(|X-3| \le .5) = 1 - P(2.5 \le X \le 3.5) =$
$1 - \int_{-.5}^{.5} \frac{3}{4}[1-u^2]du = \cdots = 1 - .6875 = .3125$.

12.

a. $P(X < 0) = F(0) = .5$.

b. $P(-1 \le X \le 1) = F(1) - F(-1) = .6875$.

c. $P(X > .5) = 1 - P(X \le .5) = 1 - F(.5) = 1 - .6836 = .3164$.

d. $f(x) = F'(x) = \dfrac{d}{dx}\left(\dfrac{1}{2} + \dfrac{3}{32}\left(4x - \dfrac{x^3}{3}\right)\right) = 0 + \dfrac{3}{32}\left(4 - \dfrac{3x^2}{3}\right) = .09375\left(4 - x^2\right)$.

e. By definition, $F(\tilde{\mu}) = .5$. $F(0) = .5$ from **a** above, which is as desired.

14.

a. If X is uniformly distributed on the interval from A to B, then $E(X) = \int_A^B x \cdot \frac{1}{B-A} dx = \frac{A+B}{2}$, the midpoint of the interval. Also, $E(X^2) = \frac{A^2 + AB + B^2}{3}$, from which $V(X) = E(X^2) - [E(X)]^2 = \ldots = \frac{(B-A)^2}{12}$.

With $A = 7.5$ and $B = 20$, $E(X) = 13.75$ and $V(X) = 13.02$.

b. From Example 4.6, the complete cdf is $F(x) = \begin{cases} 0 & x < 7.5 \\ \frac{x-7.5}{12.5} & 7.5 \le x < 20 \\ 1 & 20 \le x \end{cases}$.

c. $P(X \le 10) = F(10) = .200$; $P(10 \le X \le 15) = F(15) - F(10) = .4$.

d. $\sigma = \sqrt{13.02} = 3.61$, so $\mu \pm \sigma = (10.14, 17.36)$. Thus, $P(\mu - \sigma \le X \le \mu + \sigma) = P(10.14 \le X \le 17.36) = F(17.36) - F(10.14) = .5776$.
Similarly, $P(\mu - 2\sigma \le X \le \mu + 2\sigma) = P(6.53 \le X \le 20.97) = 1$.

18. $f(x) = \frac{1}{1-(-1)} = \frac{1}{2}$ for $-1 \le x \le 1$

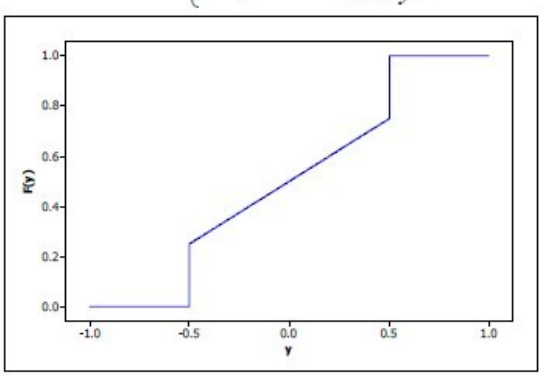
a. $P(Y = .5) = P(X \ge .5) = \int_{.5}^{1} \frac{1}{2} dx = .25$.

b. $P(Y = -.5) = .25$ as well, due to symmetry.

For $-.5 < y < .5$, $F(y) = .25 + \int_{-.5}^{y} \frac{1}{2} dx = .25 + .5(y + .5) = .5 + .5y$.

Since $Y \le .5$, $F(y) = 1$ for all $y \ge .5$. That is,

$$F(y) = \begin{cases} 0 & y < -.5 \\ .5 + .5y & -.5 \le y < .5 \\ 1 & .5 \le y \end{cases}$$



22.

a. For $1 \le x \le 2$, $F(x) = \int_1^x 2\left(1 - \frac{1}{y^2}\right)dy = 2\left(y + \frac{1}{y}\right)\Big]_1^x = 2\left(x + \frac{1}{x}\right) - 4$, so the cdf is

$$F(x) = \begin{cases} 0 & x < 1 \\ 2\left(x + \frac{1}{x}\right) - 4 & 1 \le x \le 2 \\ 1 & x > 2 \end{cases}$$

b. Set $F(x) = p$ and solve for x: $2\left(x + \frac{1}{x}\right) - 4 = p \Rightarrow 2x^2 - (p+4)x + 2 = 0 \Rightarrow$

$\eta(p) = x = \frac{(p+4) + \sqrt{(p+4)^2 - 4(2)(2)}}{2(2)} = \frac{p + 4 + \sqrt{p^2 + 8p}}{4}$. (The other root of the quadratic gives solutions outside the interval [1, 2].) To find the median $\tilde{\mu}$, set $p = .5$: $\tilde{\mu} = \eta(.5) = \ldots = 1.640$.

c. $E(X) = \int_1^2 x \cdot 2\left(1 - \frac{1}{x^2}\right)dx = 2\int_1^2 \left(x - \frac{1}{x}\right)dx = 2\left(\frac{x^2}{2} - \ln(x)\right)\Big]_1^2 = 1.614$. Similarly,

$E(X^2) = 2\int_1^2 \left(x^2 - 1\right)dx = 2\left(\frac{x^3}{3} - x\right)\Big]_1^2 = \frac{8}{3} \Rightarrow V(X) = .0626.$

d. The amount left is given by $h(x) = \max(1.5 - x, 0)$, so

$E(h(X)) = \int_1^2 \max(1.5 - x, 0) f(x)dx = 2\int_1^{1.5} (1.5 - x)\left(1 - \frac{1}{x^2}\right)dx = .061$.